

Question 1

05

For the NMOS given in the figure,

For M1 and M3 : $K_n = 0.3 \text{ mA/V}^2$ $V_{TN} = 0.6 \text{ V}$

For M2 : $K_n = 0.4 \text{ mA/V}^2$, $V_{TN} = 0.7 \text{ V}$

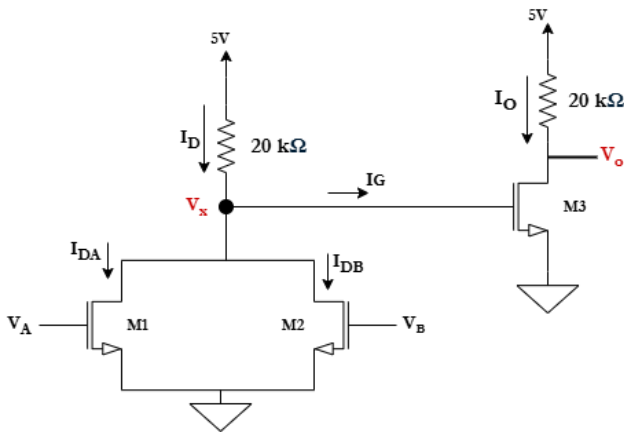


Figure 1

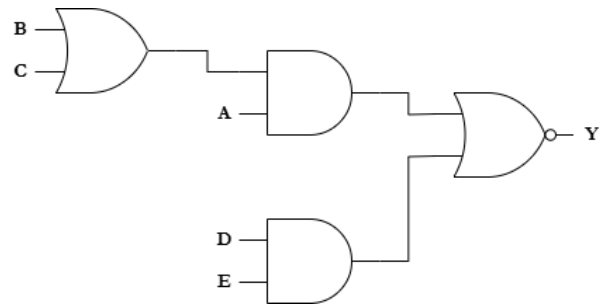


Figure 2

(a)	From the above Figure 1, determine the currents I_D , I_{DA} , I_{DB} , I_O , V_X and V_O for the cases both inputs are high and both inputs are low. (Consider 5 V as logic high and 0 V as logic low)	2
(b)	Determine the power dissipations for the cases both inputs are high and both inputs are low.	1
(c)	Find the Boolean expression for output Y from Figure 2 and using that expression design an equivalent CMOS logic circuit of Figure 2.	2

Solution:

a) Both inputs are **HIGH**, so assume M1 and M2 are in triode mode.

$$I_D = I_{DA} + I_{DB}$$

$$(5 - V_X)/20 = 0.3 [2(5 - 0.6)V_X - V_X^2] + 0.4 [2(5 - 0.7)V_X - V_X^2]$$

$$> -14V_X^2 + 122.6V_X - 5 = 0$$

$$V_X = 8.72, 0.041$$

$$\text{So } V_X = 0.041 \text{ V}$$

$$I_D = 0.248 \text{ mA}$$

$$I_{DA} = 0.3 [2(5 - 0.6)(0.041) - (0.041)^2] = 0.1077 \text{ mA}$$

$$I_{DB} = 0.4 [2(5 - 0.7)(0.041) - (0.041)^2] = 0.1404 \text{ mA}$$

Since $V_X = V_{GM3} < 0.6 \text{ V}$, thus M3 is in cutoff mode

$$I_O = 0 \text{ mA}$$

$$I_G = 0 \text{ mA}$$

$$V_O = 5 \text{ V}$$

Both inputs are **LOW**, so M1 and M2 are in cutoff mode since $V_G < V_{TN}$

$$I_D = I_{DA} = I_{DB} = 0 \text{ mA}$$

$$\text{So } V_X = 5 \text{ V}$$

$$I_G = 0 \text{ mA}$$

Since $V_X = V_{GM3} = 5 \text{ V}$, thus M3 is assumed to be in triode mode

$$I_O = 0.3 [2(5 - 0.6)V_O - V_O^2]$$

$$> (5 - V_O) / 20 = 0.3 [2(5 - 0.6)V_O - V_O^2]$$

$$> -6V_O^2 + 53.8V_O - 5 = 0$$

$$V_O = 8.87 \text{ V}, 0.094 \text{ V}$$

$$\text{So, } V_O = 0.094 \text{ V}$$

$$I_O = 0.2453 \text{ mA}$$

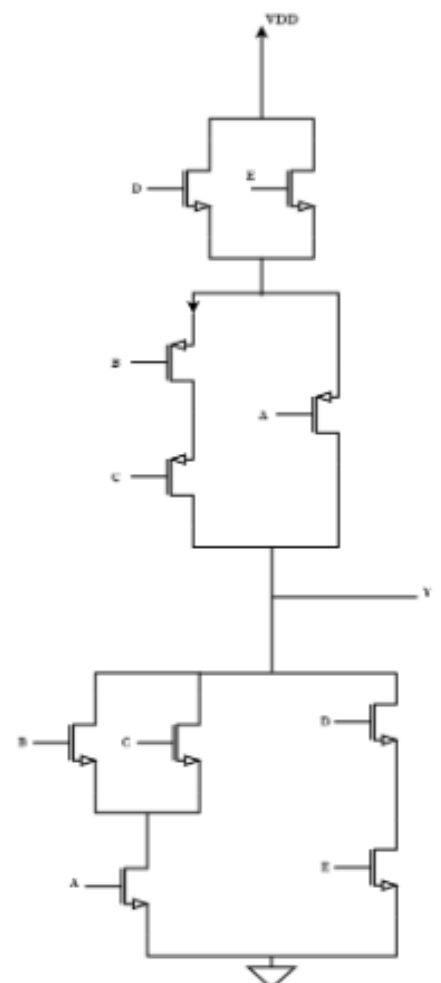
b) Power dissipation for Case (0,0)

$$P = (5 - 0) \cdot 0.2453 = 1.2265 \text{ mW}$$

Power dissipation for Case (1,1)

$$P = (5 - 0) \cdot 0.248 = 1.24 \text{ mW}$$

$$\text{c) } f = \overline{(B + C) \cdot A + D \cdot E}$$



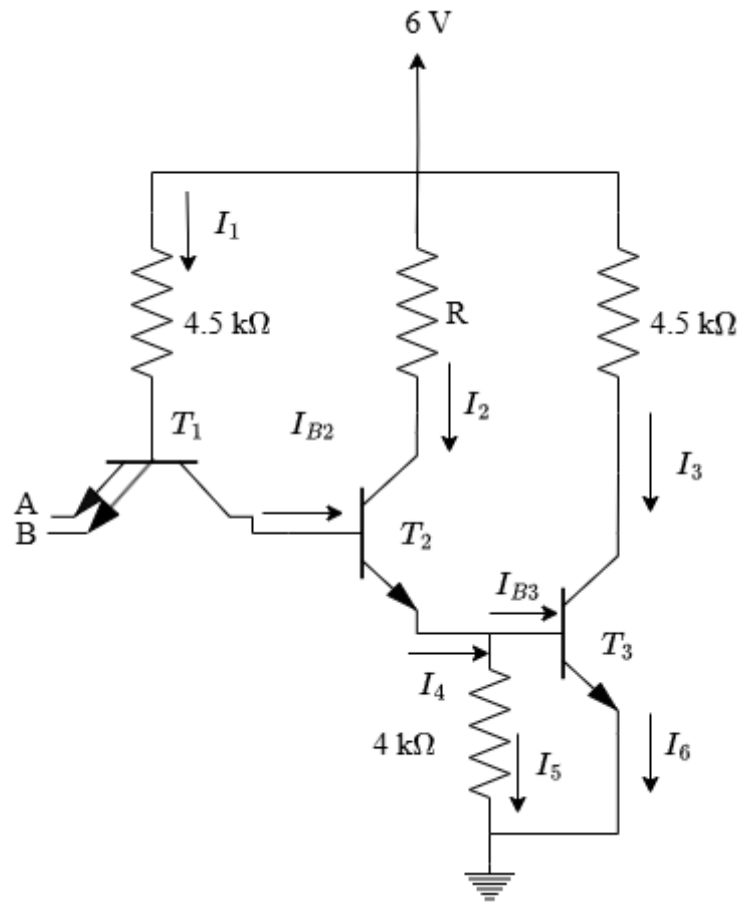
Question 2

05

For the given TTL NAND gate,

$\beta_R = 0.2$, $V_{CE}(\text{saturation}) = 0.2 \text{ V}$, $V_{BE}(\text{saturation}) = 0.8 \text{ V}$, $V_{BC}(\text{reverse active}) = 0.7 \text{ V}$

$V_{in}(\text{high}) = 6 \text{ V}$, $V_{in}(\text{low}) = 0.2 \text{ V}$



(a)	If $I_6 = 4.74 \text{ mA}$ when both inputs are high , find the value of R .	[5]
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Solution:

$$a) \quad I_6 = I_3 + I_{B3} = I_3 + I_4 - I_5 = I_3 + I_{B2} + I_2 - I_5$$

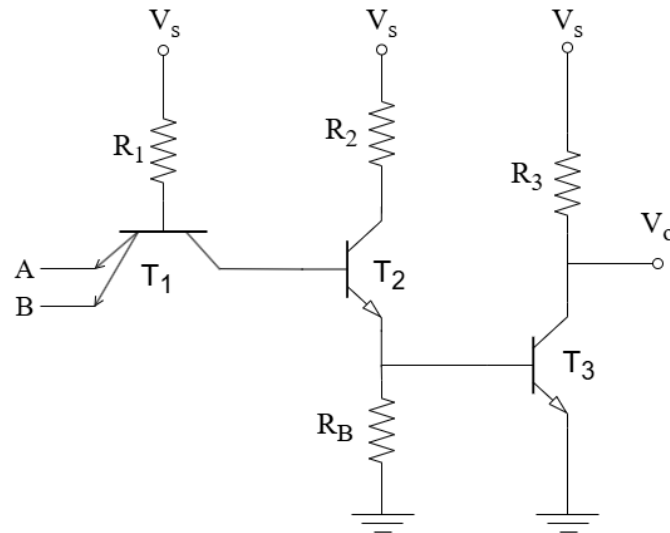
$$= I_3 + I_1 + I_{EA} + I_{EB} + I_2 - I_5$$

$$\rightarrow 4.74 = \frac{6-0.2}{4.5} + \frac{(6-2.3)}{4.5} + 0.2 \times \frac{6-2.3}{4.5} \times 2 + \frac{6-1}{R} - \frac{0.8}{4}$$

$$R = 2 \text{ k}\Omega$$

Question 3

05



$V_s = 9\text{ V}$, $R_2 = 2\text{ k}$, $R_3 = 4.4\text{ k}$, $R_B = 4.4\text{ k}$

(a)	Determine the power dissipation of this TTL circuit when both inputs are HIGH. You must show your calculation. Assume $R_1 = 6.7\text{ k}$ for this question only	3
(b)	Suppose you are selling this TTL logic gate to a customer and he states a strict criteria of keeping the power dissipation below 1.23 mW when both inputs are LOW (0 V). Determine the maximum value of R_1 you can use to satisfy this criteria.	2

Solution:

a. $I_{B1} = V_S - 2.3R1 = 1 \text{ mA}$

$$I_{E1a} = I_{E1b} = \beta_R * I_{B1} = 0.1 * 1 = 0.1 \text{ mA}$$

$$I_{C1} = I_{B2} = I_{B1} + I_{E1a} + I_{E1b} = 1.2 \text{ mA}$$

$$I_{C2} = V_S - 1R2 = 4 \text{ mA}$$

$$I_{E2} = I_{C2} + I_{B2} = 4 + 1.2 = 5.2 \text{ mA}$$

$$I_{RB} = 0.8R_B = 0.2 \text{ mA}$$

$$I_{B3} = I_{E2} - I_{RB} = 5 \text{ mA}$$

$$I_{C3} = V_S - 0.2R3 = 2 \text{ mA}$$

$$I_{E3} = I_{C3} + I_{B3} = 2 + 5 = 7 \text{ mA}$$

Power Dissipation

$$= V * I$$

$$= (V_S - 0) (I_{RB} + I_{E3})$$

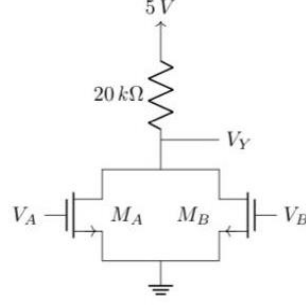
$$\Rightarrow \mathbf{64.8 \text{ mW}}$$

b. Power Dissipation = $V * I$

$$PD = (V_S - 0) * V_S - 0.8R1$$

$$R1 = V_S * (V_S - 0.8)PD$$

$$R1 = 60 \text{ k}\Omega$$



For the NOR gate shown, $V_{TN_A} = 0.25\text{ V}$, $V_{TN_B} = 0.3\text{ V}$, $K_{n_A} = 0.4 \frac{\text{A}}{\text{V}^2}$, $K_{n_B} = 0.5 \frac{\text{A}}{\text{V}^2}$.

(a)	Find the lower threshold of output voltage.
(b)	Find the average power dissipation.
(c)	If a new NMOS M_C is used instead of M_B such that the drain currents of M_A & M_C are equal when both inputs are high, find the value of K_{n_C} . It is given that $V_{TN_C} = V_{TN_B} = 0.3\text{ V}$.

Solution:

(a) Case (0,1): M_A in cut-off, M_B can be assumed to be in triode. So,

$$\begin{aligned}
 I_{20} &= I_{DS_B} \\
 \Rightarrow \frac{5 - V_Y}{20} &= K_{n_B} [2(5 - V_{TN_B})V_Y - V_Y^2] \\
 \Rightarrow \frac{5 - V_Y}{20} &= 0.5 [2(5 - 0.3)V_Y - V_Y^2] \\
 \Rightarrow V_Y &= 9.45\text{ V}, 0.052\text{ V}
 \end{aligned}$$

9.45 V is greater than 5 V, so unacceptable. Thus, $V_Y = 0.052\text{ V}$. This aligns with our assumption of triode mode as $V_{OV} = 5 - 0.3 = 4.7\text{ V}$.

Case (1,0): M_A can be assumed to be in triode, M_B will be off. So,

$$\begin{aligned}
 I_{20} &= I_{DS_A} \\
 \Rightarrow \frac{5 - V_Y}{20} &= K_{n_A} [2(5 - V_{TN_A})V_Y - V_Y^2] \\
 \Rightarrow \frac{5 - V_Y}{20} &= 0.4 [2(5 - 0.25)V_Y - V_Y^2] \\
 \Rightarrow V_Y &= 9.56\text{ V}, 0.064\text{ V}
 \end{aligned}$$

For the same reasons, 9.56 is unacceptable. So, $V_Y = 0.064\text{ V}$.

Case (1,1): Both M_A & M_B can be assumed to be in triode mode. So,

$$\begin{aligned}
 I_{20} &= I_{DS_A} + I_{DS_B} \\
 \Rightarrow \frac{5 - V_Y}{20} &= K_{n_A} [2(5 - V_{TN_A})V_Y - V_Y^2] + K_{n_B} [2(5 - V_{TN_B})V_Y - V_Y^2] \\
 \Rightarrow \frac{5 - V_Y}{20} &= 0.4 [2(5 - 0.25)V_Y - V_Y^2] + 0.5 [2(5 - 0.3)V_Y - V_Y^2] \\
 \Rightarrow V_Y &= 9.47\text{ V}, 0.029\text{ V}
 \end{aligned}$$

For the same reasons, 9.47 V is unacceptable. So, $V_Y = 0.029\text{ V}$.

$$\therefore V_{OL} = \max(0.052, 0.064, 0.029) = 0.064\text{ V}$$

(b) For case (0,0), $I_{20} = I_{DS_A} = I_{DS_B} = 0$. So,

$$P_{(0,1)} = (5 - 0) \times \frac{5 - 0.052}{20} = 1.237 \text{ mW}$$

$$P_{(1,0)} = (5 - 0) \times \frac{5 - 0.064}{20} = 1.234 \text{ mW}$$

$$P_{(1,1)} = (5 - 0) \times \frac{5 - 0.029}{20} = 1.24275 \text{ mW}$$

$$P_{(0,0)} = (5 - 0) \times 0 = 0 \text{ mW}$$

$$\therefore P_{avg} = \frac{1.237 + 1.234 + 1.24275 + 0}{4} = 0.93 \text{ mW}$$

(c) It is given that,

$$I_{DS_A} = I_{DS_C}$$

$$\Rightarrow K_{n_A} [2(5 - V_{TN_A})V_Y - V_Y^2] = K_{n_C} [2(5 - V_{TN_C})V_Y - V_Y^2]$$

$$\Rightarrow 0.4[2(5 - 0.25)V_Y - V_Y^2] = K_{n_C} [2(5 - 0.3)V_Y - V_Y^2]$$

Now,

$$I_{20} = I_{DS_A} + I_{DS_C} = I_{DS_A} + I_{DS_A} = 2I_{DS_A}$$

$$\frac{5 - V_Y}{20} = 2 \times 0.4[2(5 - 0.25)V_Y - V_Y^2]$$

$$\Rightarrow V_Y = 9.53 \text{ V}, 0.0325 \text{ V}$$

For the same reasons, $V_Y = 0.0325 \text{ V}$. Using this value we get,

$$\Rightarrow 0.4[2(5 - 0.25)0.0325 - 0.0325^2] = K_{n_C} [2(5 - 0.3)0.0325 - 0.0325^2]$$

$$\Rightarrow K_{n_C} = 0.404 \frac{A}{V^2}$$